

[Total No. of Questions - 9] [Total No. of Printed Pages - 2]

Dec-24-0342 (NEP)

MCA-7302 (Analysis and Design of Algorithms)

MCA-3rd CBCS/NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

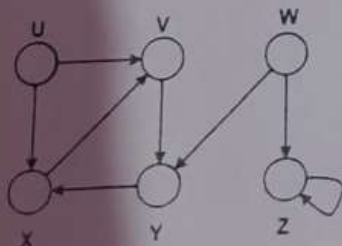
Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

SECTION-A

1. Define asymptotic notations. Explain them with suitable examples. (12)
2. What is Divide and Conquer approach? Write an algorithm for Merge Sort with its time and space complexity. (12)

SECTION-B

3. What is the main aim of lower bound theory? Draw comparison tree for binary search by taking suitable example and find the number of comparisons required to perform a task of the search of n terms. (12)
4. Explain the concept of Depth-First Search (DFS) Graph traversal technique with its time and space complexity. Traverse following given graph $G(V,E)$ using DFS. (Start traversal from Vertex U). (12)



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SECTION-C

5. (a) Differentiate between P and NP problems. (6)
- (b) Explain polynomial reduction with example. (8)
6. Explain the working of randomized quick sort by taking suitable example. Compare the time complexity of randomized quick sort with simple quick sort. (12)

SECTION-D

7. Given text string with length n and a pattern with length m , write a naive algorithm to prints all occurrences of pattern in text. (12)
8. Discuss string matching algorithm with finite automata in detail. (12)

SECTION-E (Compulsory)

9. (a) How do branch and bound methodology works? (2)
- (b) Differentiate between Kruskal's and Prim's algorithm for Minimum Spanning Tree. (2)
- (c) Briefly explain the role of BFS in finding the shortest path in an unweighted graph. (2)
- (d) Define NP-hard problems. (2)
- (e) Describe the basic concept of a randomized algorithm and give one example where such an algorithm might be used. (2)
- (f) Compare the time complexity of various string matching algorithms. (2)